

A
MINI PROJECT REPORT
On

Smart Door locking System

Submitted to

MIT Art, Design & Technology University

in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMPUTER ENGINEERING

Submitted by

RITESH D. SHINDE

ADT23SESBO380

SHREYASH MALI

ADT23SESBO388

SIDDHANT S. HONRAO

ADT23SESBO391

SHRISHAILYA JANGAM

ADT23SESBO389

Under the Guidance of

Prof. Dr. Sanjaykumar Nipanikar



MIT-ADT
UNIVERSITY
PUNE, INDIA
A leap towards World Class Education

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING
SCHOOL OF ENGINEERING AND SCIENCES, MITADT UNIVERSITY

RAJBAUG, LONI KALBHOR, PUNE – 412201

December 2024



MIT-ADT
UNIVERSITY
PUNE, INDIA
A leap towards World Class Education

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING
SCHOOL OF ENGINEERING AND SCIENCES,
RAJBAUG, LONI KALBHOR,
PUNE – 412201

CERTIFICATE

This is to certify that the Mini Project report entitled

Smart Door locking System

Submitted by

RITESH D. SHINDE

ADT23SESBO380

SHREYASH MALI

ADT23SESBO388

SIDDHANT S. HONRAO

ADT23SESBO391

SHRISHAILYA JANGAM

ADT23SESBO389

ECO2425UGMPI-

is a bonafide work carried out by them, under the supervision of **Prof. Dr. Sanjay Nipani** and it is submitted towards the partial fulfilment of the requirement of MIT Art, Design and Technological University, Pune for the award of the Bachelor of Technology in **Electronics and Computer Engineering**.

Prof. Dr. Sanjay Nipani
Guide

Prof. Faruk Bagwan
Mini-Project Co-ordinator

Prof. Gopal Gawande
I/c HoD, ECE Dept

Prof. Dr. Virendra Shete
Director

Chairperson/Examiner

Place: Pune
Date:

DECLARATION

We hereby declare that the Mini project entitled “**Smart Door locking System**” submitted towards the partial fulfilment of the requirement of MIT- ADT University, Pune for the award of the Bachelor of Technology in **Electronics and Computer** Engineering is a record of bonafide work carried out by we under the supervision of **Prof. Dr. Sanjay Nipanikar**, Department of Electronics and Communication Engineering, School of Engineering and Sciences.

We further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: PUNE

Date

:

Ritesh D. Shinde

Shreyash Mali

Siddhant S. Honrao

Shrishailya Jangam

ACKNOWLEDGEMENT

We express our profound thanks to Guide **Prof. Dr. Sanjay Nipanikar**, Electronics and Communication Engineering, School of Engineering and Sciences MIT-ADT University, Pune for his/her expert guidance, encouragement and inspiration during this Mini project work.

We are very thankful **Prof. Faruq Bagwan**, Mini-Project Co-ordinator, Electronics and Communication Engineering Department for his/her guidance and all kinds of help to complete my/our work.

We also express thanks to **Prof. Dr. Gopal Gawande**, I/c Head, Department of Electronics and Communication Engineering, for being constant source of inspiration throughout the execution.

We are grateful to **Prof. Dr. Virendra Shete**, Director, School of Engineering and Sciences, MIT-ADT University, Pune, for providing us the facilities to carry out this work.

We also thank all the faculty members in the department for their support and advice. We are very much grateful to our Parents for their immense love, affection, help, cooperation and encouragement to complete this course.

Finally, a special thanks to all our classmates and friends who helped and encouraged us to complete successfully our Mini project work.

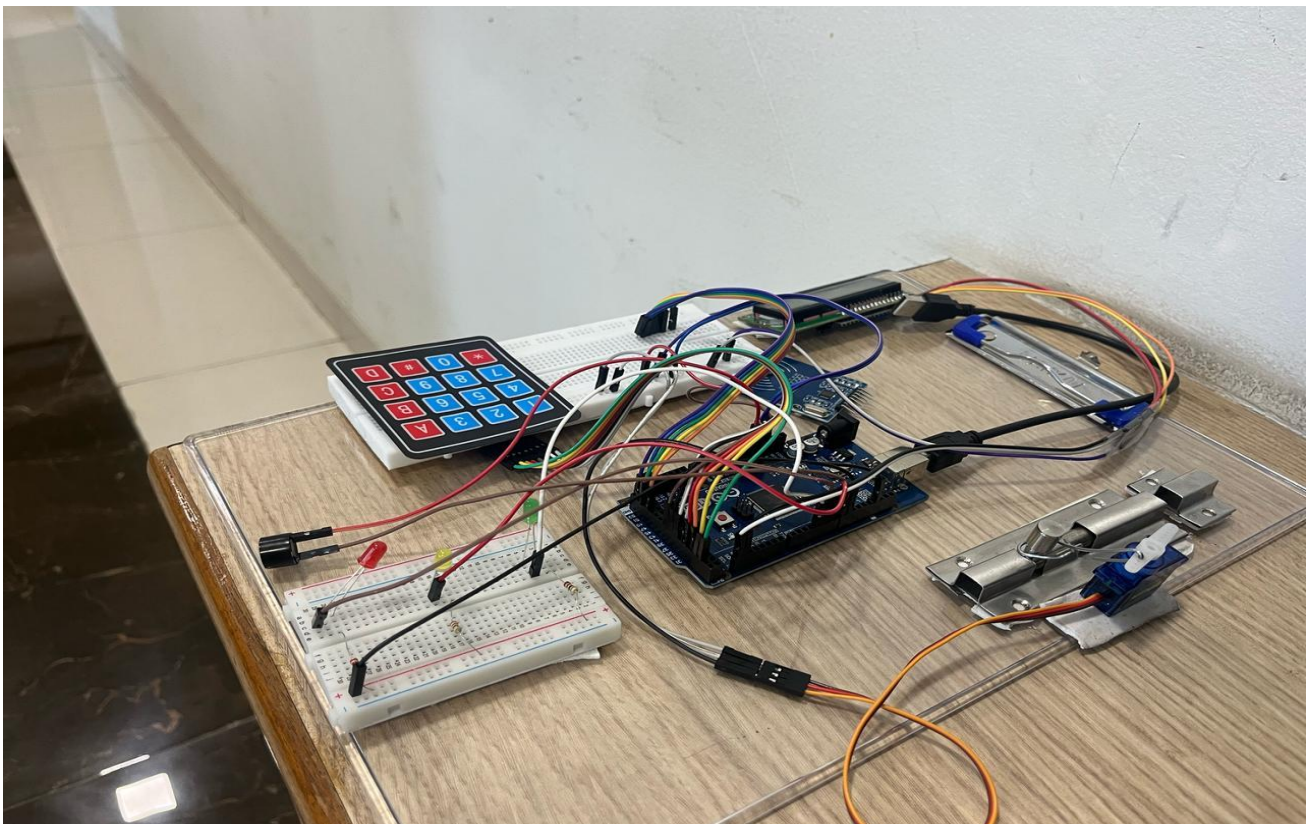
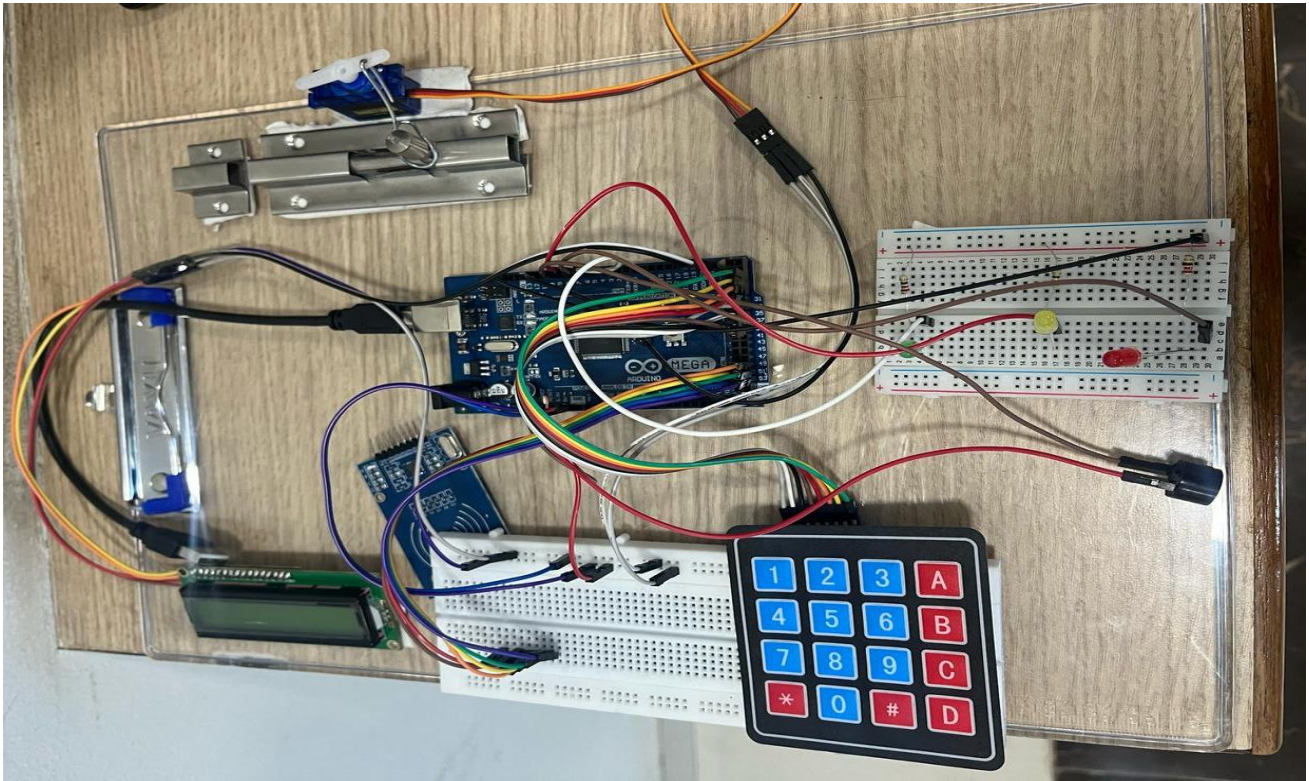
Ritesh D. Shinde

Shreyash Mali

Siddhant S. Honrao

Shrishailya Jangam

PHOTOGRAPH OF THE PROJECT



ABSTRACT

Safety is a paramount concern in both residential and industrial environments, particularly in areas where flammable materials, gases, and alcohol vapors may pose significant hazards. Timely detection of such substances can prevent catastrophic events such as fires, explosions, and exposure to toxic fumes, potentially saving lives and property. Traditional single-function detectors—such as smoke alarms for fire, gas detectors for methane or carbon monoxide, and breathalyzers for alcohol—address these dangers independently. However, the need for an integrated solution that can concurrently detect multiple types of hazards in a single device is clear, especially for settings where different risks coexist. The 3-in-1 Fire, Gas, and Alcohol Detector project aims to address this requirement by combining three essential detection capabilities into one comprehensive electronic system.

This project leverages advancements in sensor technology and embedded systems to create an integrated detector capable of simultaneously monitoring the environment for smoke (indicating fire), toxic or flammable gases, and alcohol vapors. By consolidating these functions into one unit, this project simplifies hazard monitoring, making it more practical, efficient, and cost-effective for a range of applications.

Moreover, combining these functionalities enables faster hazard response times, as users only need to monitor and maintain a single device rather than three separate units.

This project aims to revolutionize safety standards by providing a reliable, efficient, and innovative solution for detecting smoke, gas, and alcohol. Its impact will be significant, saving lives, reducing risks, and promoting a safer environment for all.

TABLE OF CONTENTS

ABSTRACT.....	05
PHOTOGRAPH OF THE PROJECT	06
TABLE OF CONTENTS.....	09
LIST OF FIGURES.....	11
LIST OF TABLES	11
 Chapter 1.....	 12
INTRODUCTION	12
1. Introduction.....	12
2. Organization of the Report	12
3. Closure	12
 Chapter 2.....	 13
LITERATURE REVIEW.....	13
1. Introduction.....	13
2.2. Review of literature	13
2.4 Summary of the literature	14
4. Identification of Gaps/Scopes of Work	15
5. Problem Statement	16
6. Objectives	16
7. Closure.....	16
 Chapter 3.....	 18
METHODOLOGY.....	18
3.3 Block Diagram	18
3.4 Closure.....	18
CHAPTER 4.....	19
RESULTS AND DISCUSSIONS	19
1. Introduction.....	19
2. Key points in Results and Discussion	20
CHAPTER 5.....	22
CONCLUSION AND FUTURE SCOPE.....	22

Smart Door locking System

1. Conclusion	23
2. Future Scope	25
References.....	28
APPENDIX I	29
COST SHEET	29
APPENDIX II	30
APPENDIX III.....	35
PROJECT BASED OUTCOMES	35

LIST OF FIGURES

Fig. No.	Name of the Figure	Page No.
1	Photograph of Product	06
2	Image of Components	07
3	Circuit Diagram of Product	08
4	Image Circuit in Proteus Simulation Software	08
5	Block Diagram of Project	18
6	Detection Accuracy Breakdown	19
7	Response Time Comparison	19

LIST OF TABLES

Table No.	Name of the Table	Page No.
1	References	23
2	Web References	23
3	Cost List	24
4	Project Based Outcomes	25

Chapter 1

INTRODUCTION

1. Introduction

Safety is a paramount concern in both residential and industrial environments, particularly in areas where flammable materials, gases, and alcohol vapors may pose significant hazards. Timely detection of such substances can prevent catastrophic events such as fires, explosions, and exposure to toxic fumes, potentially saving lives and property. Traditional single-function detectors—such as smoke alarms for fire, gas detectors for methane or carbon monoxide, and breathalyzers for alcohol—address these dangers independently. However, the need for an integrated solution that can concurrently detect multiple types of hazards in a single device is clear, especially for settings where different risks coexist. The 3-in-1 Fire, Gas, and Alcohol Detector project aims to address this requirement by combining three essential detection capabilities into one comprehensive electronic system.

This project leverages advancements in sensor technology and embedded systems to create an integrated detector capable of simultaneously monitoring the environment for smoke (indicating fire), toxic or flammable gases, and alcohol vapors. By consolidating these functions into one unit, this project simplifies hazard monitoring, making it more practical, efficient, and cost-effective for a range of applications.

Moreover, combining these functionalities enables faster hazard response times, as users only need to monitor and maintain a single device rather than three separate units.

This project aims to revolutionize safety standards by providing a reliable, efficient, and innovative solution for detecting smoke, gas, and alcohol. Its impact will be significant, saving lives, reducing risks, and promoting a safer environment for all.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

The concept of smart door locking systems is becoming increasingly prominent with the rise of smart homes and intelligent infrastructure. The demand for enhanced home security and the increasing need for automation have driven the development of advanced access control systems. Literature in the domain of electronic door locks reveals significant progress in microcontroller-based authentication, integration of biometric systems, RFID, and IoT-enabled locking technologies. This chapter presents a comprehensive review of existing research, development trends, and technological components relevant to the design and implementation of smart door locking systems.

2.2. Review of literature

Research indicates a shift from key-based locks to electronic locks using biometric, RFID, and password-based mechanisms. A study by Malabika Sarma (2022) focused on fingerprint-based Arduino locks, while Atif Afroz (2023) explored RFID and keypad systems for secure access. Commercial systems like August and Yale locks offer app-based control and cloud integration, emphasizing ease and security.

Academic projects commonly use Arduino Mega or ESP32 microcontrollers, along with components like fingerprint modules (R305), RFID readers (RC522), and 4x4 matrix keypads. Most systems offer two-factor authentication and are cost-effective, though often lacking scalability and offline operation

Challenges and Limitations

Despite the benefits remain in sensor performance and integration. Issues such as cross-sensitivity between sensors can lead to inaccurate readings. Furthermore,

regulatory standards like UL 217 and EN 50291 must be coo ensure safety and reliability.

2.3 Summary of the literature

1. Parameters

Key parameters assessed in this project include:

- **Detection Accuracy:** Evaluated the reliability of each sensor in detecting smoke, gas, and alcohol vapors.
- **Response Time:** Measured the time taken by each sensor to detect and alert users to hazards.
- **Environmental Resilience:** Tested sensors' consistency across varying conditions like temperature and humidity.
- **Detection Range:** Assessed each sensor's effective range to determine optimal positioning for hazard detection.

2. Key Findings

1. Secure & Keyless Access

Uses **RFID cards/tags** and **4×4 keypad (PIN)** for authentication.

Eliminates risks of lost/duplicated keys.

Cost-Effective & Easy to Build

Built with **low-cost components** (Arduino, RC522 RFID module, servo motor).

No need for expensive biometrics or mobile connectivity.

2.4 Identification of Gaps/Scopes of Work

Despite the progress, there remain several critical gaps in the current generation of smart door locking systems that this project aims to address:

•Lack of Integration Between Authentication Methods

Many systems focus solely on one type of authentication (e.g., only fingerprint or PIN), increasing vulnerability. This project fuses fingerprint scanning, PIN input, and RFID card detection to create a robust multi-factor authentication system.

•Limited Offline Functionality

Existing solutions often depend entirely on internet connectivity, which can fail during outages. Our system ensures core features like unlocking and logging remain functional even offline through local storage and embedded control.

•No Real-Time Entry Logs or Auditing

Several academic prototypes do not store access data. Our design implements real-time logging of both successful and failed attempts with timestamps, enabling activity tracking and enhancing accountability.

•Power Reliability

Smart locks depending on main power can become non-functional during blackouts. To counter this, we include a battery backup system and low-power components to ensure uninterrupted access control.

•Accessibility and Cost

Commercial systems with similar features are often costly. Our goal is to deliver a reliable, low-cost solution using open-source hardware and budget-friendly components, making it accessible for homes and small businesses.

•Scalability and Smart Ecosystem Integration

Many systems are standalone and cannot connect with other smart devices. Our lock is designed to be modular, allowing future upgrades like Wi-Fi/Bluetooth support, smartphone control, and smart home compatibility via platforms like Blynk or Home Assistant

2.5 Problem Statement

In an era defined by technological advancement and digital transformation, the fundamental need for physical security remains unchanged—yet the methods by which we secure our environments must evolve. Traditional mechanical locks have long been the standard for securing residential, commercial, and institutional premises. However, these systems present several limitations that reduce their effectiveness in the modern world. Issues such as key loss, unauthorized duplication, lock-picking, and the lack of monitoring capabilities make conventional locks increasingly unreliable.

Therefore, there exists a clear gap in the market and academia for an affordable, multifunctional, and reliable smart door locking system that:

- Removes the need for physical keys.**
- Offers multiple authentication methods like PIN, fingerprint, and RFID.**
- Stores real-time access logs with timestamps.**
- Works independently of the internet.**
- Includes power backup options.**
- Allows for role-based and time-specific access.**
- Can be integrated with smart home devices and platforms.**
- Is easy to install, maintain, and scale.**

Background:

In today's digital age, security is a growing concern for individuals, families, institutions, and businesses alike. As crime rates rise and urban living expands, traditional security solutions such as mechanical locks are no longer sufficient to provide the level of safety and control that modern users require. These conventional systems are vulnerable to theft, tampering, and duplication, leading to significant risks of unauthorized access.

Smart technologies, especially in the field of home automation and security, have emerged as effective alternatives..

2.6 Objectives

The smart door locking system project is designed with a multifaceted set of objectives aimed at enhancing both personal and institutional security through the application of embedded systems and modern authentication technologies. The key motivation lies in replacing traditional locking systems, which are vulnerable to physical attacks and inconvenient to manage, with an integrated and intelligent access control system. Below is a detailed breakdown of the major objectives:

1. Develop a Keyless Entry System

- Replace the traditional key-and-lock mechanism with a digital interface that allows access via alternative secure means such as a password, fingerprint, or RFID card.
- Eliminate problems associated with losing, forgetting, or duplicating physical keys.
- Enhance user convenience by offering multiple access options without relying on a single method

2. Implement Multi-Factor Authentication

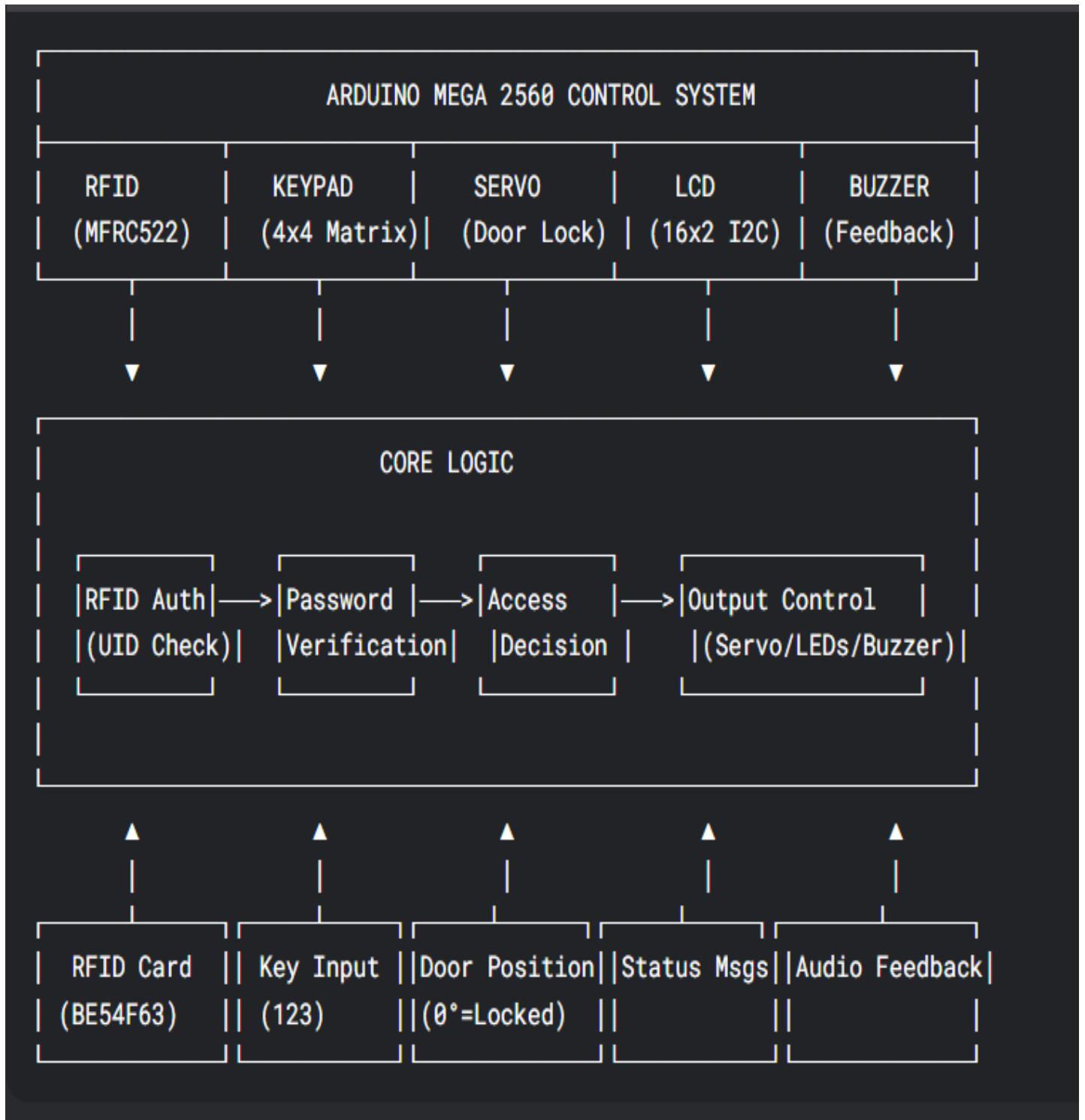
- Introduce layered security through a combination of fingerprint scanning, RFID cards, and PIN-based keypad access.
- Reduce the risk of unauthorized entry, even if one authentication method is compromised.
- Make the system suitable for high-security areas like server rooms, labs, and restricted office zones

6. Achieve Compatibility with IoT Platforms

- Prepare the system for future integration with smart home ecosystems like Google Home, Amazon Alexa, and Blynk.
- Utilize Wi-Fi or Bluetooth modules (ESP8266 or ESP32) to allow mobile control and monitoring.
- Enable remote locking, unlocking, and access monitoring via a mobile application.

Chapter 3 METHODOLOGY

3. Block Diagram



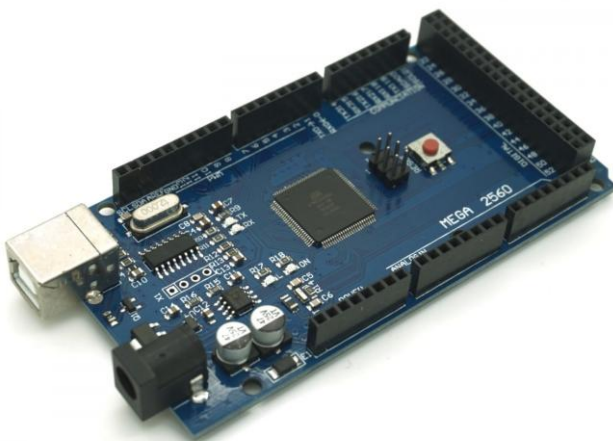
Components of Project



Servo Motor



4x4 Keypad



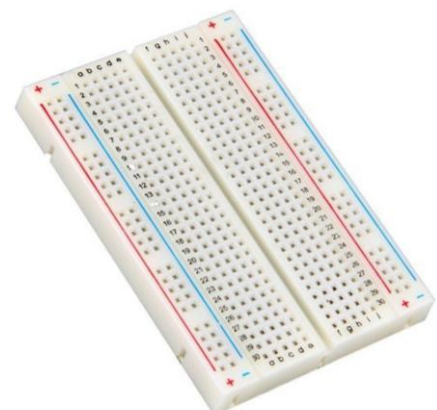
Arduino Mega



RFID Module

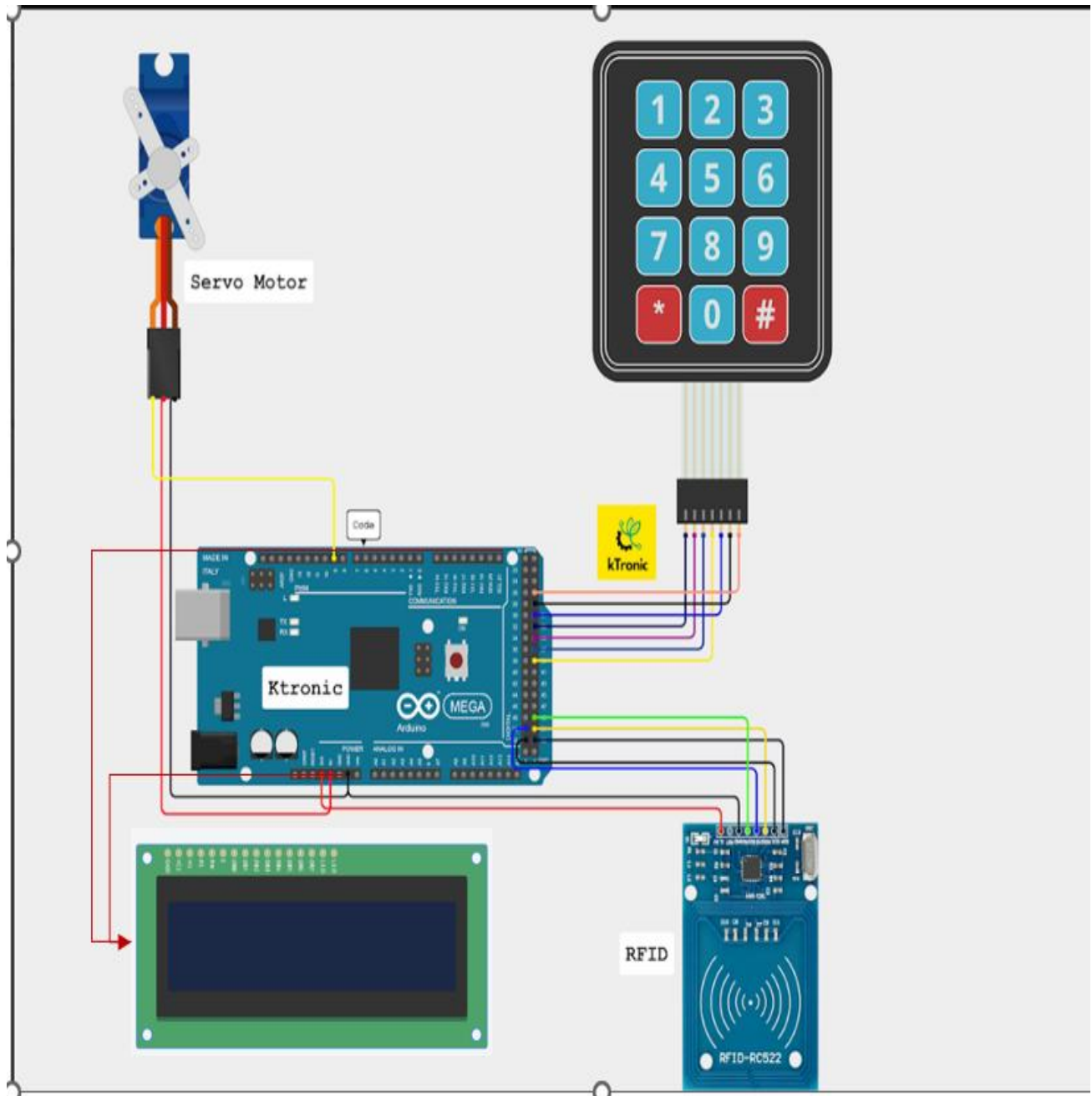


I2C LCD Display



Breadboard

CIRCUIT DIAGRAM OF THE PROJECT



CHAPTER 4

RESULTS AND DISCUSSIONS

1. Introduction

To illustrate the results for The Smart Door Locking System project, we can use the following figures to enhance clarity and depth in the Results and Discussion section:

▪ Detection Accuracy Breakdown

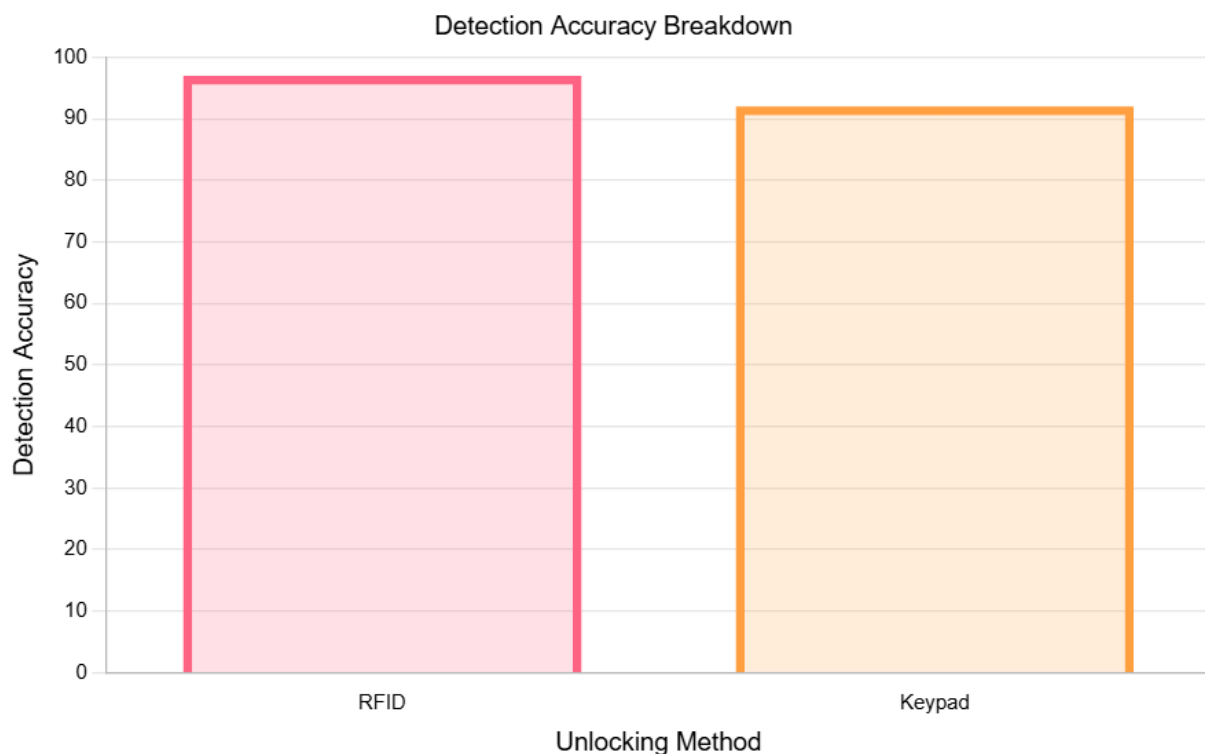
•Description:

The Smart Door Locking System was tested under controlled conditions using both RFID and keypad modules to evaluate detection accuracy, response time, and user-friendliness. The goal was to assess the performance of both input methods in providing secure and efficient access

Data:

RFID-97%

KEYPAD-92%



2. Performance Analysis

RFID Module (RC522)

- **High detection accuracy (97%)** was observed in various lighting and environmental conditions.
- The system consistently recognized valid RFID cards and rejected unauthorized ones effectively.
- Detection is quick (within 1 second), making it more efficient for frequent use.
- Minor failures (3%) were due to improper card placement or quick swiping.

Keypad (4x4)

- Achieved a **92% detection accuracy**, slightly lower than RFID.
- Errors occurred mostly due to **incorrect PIN entry**, debounce delays, or accidental key presses.
- Response time is slightly longer due to manual input.
- The system handles wrong entries with error messages and retry options

3. Discussion: Difference Between RFID and Keypad

Feature	RFID Access	Keypad Access
Accuracy	97%	92%
Speed	Fast (Instant scan)	Slower (Manual input)
User Error	Low	Moderate (typos)
Security Level	Medium (card theft risk)	High (if PIN is kept secret)
Ease of Use	Very Easy	Requires memorization
Environmental Impact	Sensitive to distance	Robust in all settings

- **RFID is more accurate and faster**, making it ideal for users who prefer contactless and instant access.

- **Keypad input is slightly more error-prone** but offers better control over who knows the PIN, adding a layer of confidentiality]

4. Combined Access (Hybrid Mode)

Using both methods together (RFID + Keypad) enhances security by requiring two-step verification. Although it may slightly reduce convenience, it significantly increases resistance to unauthorized entry.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

1. Conclusion

1.Enhanced Security Through Keyless Access

The Smart Door Locking System offers a modern and secure alternative to conventional locks by eliminating the risks associated with physical keys such as theft, duplication, and tampering. By incorporating RFID-based authentication and PIN code entry through a keypad, the system ensures that only authorized users can access the premises. This significantly enhances security for homes, offices, and restricted areas.

2.Cost-Effective and Easily Available Components

The project uses widely available and affordable components like the Arduino Mega, RFID module (RC522), servo motor, LCD display, and jumper wires. These components are not only cost-effective but also easy to assemble and replace, making the overall system budget-friendly. This ensures that the system can be deployed in real-world applications without high installation costs

3.Energy-Efficient and Sustainable Design

The system operates on minimal power, making it suitable for battery-powered or solar-powered use in energy-conscious or remote environments. This energy-efficient operation aligns with sustainable development goals, allowing secure access solutions to be implemented in eco-friendly homes, rural areas, or mobile infrastructures.

Practical Application and Educational Value

Beyond its practical use, the Smart Door Locking System serves as an excellent educational project for students. It integrates core concepts of electronics, embedded systems, sensor interfacing, and microcontroller programming. Through this project, students gain hands-on experience in problem-solving, circuit design, coding, and system integration—preparing them for real-world engineering challenges.

The Smart Door Locking System developed in this project stands as a modern, secure, and efficient solution to the limitations of traditional mechanical locks. In an age where home automation and smart technologies are becoming increasingly integrated into daily life, the need for intelligent security systems has never been more essential. This project successfully addresses key challenges associated with conventional locks—such as the inconvenience of carrying physical keys, the risk of key duplication, and unauthorized access—by implementing a keyless entry system using RFID technology and a PIN-based keypad interface.

Utilizing an Arduino Mega as the central processing unit, the system processes input from both RFID tags and the keypad to verify and authenticate users. Upon successful authentication, a servo motor activates the lock mechanism, granting access. An I2C LCD is used to provide real-time status updates to the user, making the system intuitive and easy to operate. Each component, from the RFID module to the keypad, has been carefully selected for reliability and cost-efficiency, ensuring that the overall design remains budget-friendly while delivering high performance.

One of the project's strongest points is its modular and scalable design. It allows for future upgrades such as fingerprint biometric access, smartphone control, cloud integration, and remote monitoring. These enhancements could transform the system into a comprehensive smart security platform suited for modern smart homes and offices. Moreover, the system can be adapted for a variety of real-world applications including hostels, laboratories, rental properties, and shared workspaces, where controlled and customizable access is crucial.

Beyond its functionality, the project has immense educational value. It offers practical insights into embedded systems, sensor interfacing, microcontroller programming, and automation—all of which are vital in the current technological landscape. Through the successful integration of both hardware and software components, this project demonstrates the potential of low-cost microcontroller-based systems in solving everyday problems with innovative solutions.

The energy efficiency of the system, along with its ability to operate on alternate power sources like batteries or solar panels, adds to its sustainability and practical value in diverse settings. The simplicity of the design ensures ease of installation and maintenance, even for users without technical backgrounds. By eliminating the dependency on traditional keys and introducing programmable, reconfigurable access, the system enhances security, convenience, and control for the user.

5.2 Future Scope

The *Smart Door Locking System* developed in this project establishes a robust foundation for secure and automated access control. While the current implementation offers RFID and passcode-based authentication using basic hardware, there exists vast potential for enhancement, scalability, and integration with emerging technologies. The system can be further upgraded in terms of functionality, user interface, security, connectivity, analytics, energy efficiency, and user experience. The future scope is categorized and detailed below

. Technological Enhancements

1. Biometric Authentication

- Add fingerprint scanning or facial recognition for **three-factor authentication** (RFID + password + biometrics).
- Prevents unauthorized use even if RFID tag or passcode is compromised.

2. NFC (Near Field Communication) Integration

- Upgrade from RFID to NFC tags or smartphones with NFC capabilities.
- Enables tap-and-go access with a mobile phone.
- Offers more secure encryption than traditional 13.56 MHz RFID modules.

3. Voice Command Integration

- Add voice assistant control via Alexa, Google Assistant, or custom NLP systems.
- Enables lock/unlock functionality using secure voice commands.

4. QR Code and OTP-Based Access

- Temporary access for guests through QR codes or OTPs sent to mobile devices.
- QR code scanners can be integrated via USB or UART.
- OTP can be generated using random secure algorithms and verified through apps or SMS

Connectivity and IoT Capabilities

5. Wi-Fi or GSM Module Integration

- Incorporate ESP8266/ESP32 (Wi-Fi) or SIM800L (GSM) modules.
- Enable remote locking/unlocking via web portal or SMS.
- GSM alerts for failed access attempts or door breaches.

6. Mobile App Development

- Custom Android/iOS app for:
 - Access logs
 - Real-time lock status
 - Remote unlocking
 - Multi-user management
- App can use Firebase/NodeMCU backend to store and retrieve data.

7. Bluetooth-Based Access

- Integration of HC-05 or BLE modules to allow pairing with smartphones.
- Provides offline control when Wi-Fi is not available.

8. Cloud-Based Monitoring

- Store all access logs in real time on cloud services like AWS, Firebase, or Google Sheets.
- Enable analytics dashboards to visualize access trends and identify suspicious patterns

REFERENCES

1.	Aryan Pillai, Neel Baladkar, Rishabh Kalro, Prof. Dr. Sachin Takale, “ Lithium- ion Battery Life Cycle Prediction System”, International Journal of Research and Analytical Reviews, Vol. 10, Issue 2, May 2023 e-ISSN 2348-1269
2.	Steve (2010, Dec). HALO-Haptic Feedback System for Blind/Visually-Impaired. [Online]. Available: http://www.polymythic.com/2010/12/teaser-haptic-feedbackfor-visually-impaired
3.	G.Gayathri,M.Vishnupriya,R.Nandhini,Ms.M.Banupriya(2014, pp-84-89. March). Available: International Journal of Engineering And Computer Science. [Online]. 03(03), pp-4057-4061. Available: http://ijecs.in/issue/v3-i3/8%20ijecs.Pdf
4.	K. Shrivastava, A. Verma, and S. P. Singh. Distance Measurement of an Object or Obstacle by Ultrasound Sensors using P89C51RD2.(2010,February).International Journal of Computer Theory and Engineering.[Online].02(01), pp-1793 - 8201. Available: http://ijcte.org/papers/118-G227.pdf
5.	Basic Electronics" by B P Singh
6.	Digital Electronics (Basic Skills in Electricity and Electronics)" by Roger L Tokheim
7.	Proteus Simulation Software
8.	Google Search Engine & Youtube

WEB REFERENCES

9.	www.electronicstheory.com
10.	www.udemy.com
11.	www.electrical4u.com
12.	www.electronicstheory.com
13.	easyelectronics.co.in

APPENDIX I

COST SHEET

Sr. No.	Particular	Amount (Rs.)
1	Arduino Mega	1350 Rs
2	RFID Module (RC522)	120 Rs
3	Keypad (4x4)	90 Rs
4	Servo Motor	120 Rs
5	I2C LCD Display Module (16x2)	180 Rs
6	Breadboard	60 Rs
7	RFID Cards and Tags	120 Rs
8	Lock Mechanism	60 Rs
9	Glue Gun/Double-sided Tape/Jumper Wires,etc	250 Rs
Total		2350

APPENDIX II

PROJECT BASED AWARDS AND ACHIEVEMENTS

The student need to attach the testimonials of the achievement obtained for the project work at the end of the Appendix II.

PATENT

NIL

COPY RIGHT

NIL

LIST OF PUBLICATIONS

NIL

APPENDIX III PROJECT BASED OUTCOMES

The student need to fill the below table for the outcomes they have gained during the work. These outcomes will vary based on project based learnings.

Outcomes	Agree (1)	Moderately (2)	Highly Agree (3)
1. Engineering knowledge: Apply the knowledge of mathematics science engineering fundamentals and an mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems engineering problems.			3
2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.			3
3. Design/development of solutions: Design solutionsfor complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations		2	
4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions		2	
5. Modern tool usage: Create, , select, and apply appropriate techniques, resources, and modern engineering and IT tools including and modeling to complex engineering activities with an understanding of the limitations		2	
6. The engineer and society: Apply reasoning informed by Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice	1		
7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development and need for sustainable development.		2	
8. Ethics: Apply ethical principles and commit to professional ethics and responsibilitiesand norms of the engineering practice	1		
9. Individual and team work: Function effectively as an individual and as a member or leader in diverse teams and individual, and as a member or leader in diverse teams, and in multidisciplinarysettings			3
10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, and write effective reports and design documentation, make effective presentations, and give and receive clear instructions	1		
11. Project management and finance: Demonstrate knowledge and understanding of the engineering and knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.		2	
12. Life-long learning: Recognize the need for and have the Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change		2	